STILLSTACK · FEASIBILITY ASSESSMENT

StillStack feasibility assessment

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	SSK-FEA-001	Version	0.2
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stillstack		
Source	docs/08-feasibility.md @ a65cfcd		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	PR-002 restated (D-10, decided by Amish 2026-10-25); cost USD 460.70
0.1	2026-10-24	Amish Chadha	First feasibility assessment (kit 1.9.1 upgrade)

StillStack feasibility assessment

Feasible on paper as a garage-built prototype: the working principle sits within known science (SSK-STM-001), the design is detailed to D3 and buildable with the build plan, and the open questions are material choices that a short test programme at TRL 4 closes (wick permeability, coating certificate, crevice corrosion), not the working principle. Cost is the weakest aspect: USD 460.70 against the USD 355 value-engineering target. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design (working principle)	Four-stage multi-effect diffusion still gives 10.3 L/m2/day on the design day with a gained output ratio of 1.22 to 1.29, after the independent review's corrections and the painted underside (SSK-DEC-001 D-07); independent order-of-magnitude checks agree within a factor of 1.0 to 1.15	High	Thin margin on PR-001 (3 %, at risk); PR-002 restated as 1.2 or more by Amish (D-10, 2026-10-25), met on paper with a 1.7 % margin	SSK-STM-001 S1 to S8, S27; SSK-CAL-001 sections 2 to 5
Materials	PPS rails (stages 1 to 3), silicone ribs, stone wool liner and PTFE-class coating are rated above the stagnation temperatures; stage 3 fixed by D-01	Medium	Wick fabric rating and permeability unknown (S17, S25); coating certificate not seen (S22); crevice corrosion of bare aluminium under the wicks (S21); the stainless clips on aluminium in brine are replaced by 5052 aluminium clips (S30, D-08, decided 2026-10-25); inner glazing 7 K under its limit, kept with the cover rule on the panel label (S13, D-05)	SSK-STM-001; SSK-DEC-001 D-01, D-04, D-05
Manufacturing and assembly	Plywood box, cut and folded 0.5 mm aluminium, printed rails for stage 4 only, hand tools; 22 illustrated steps; the D3 model shows every screw and bolt and found two bolt lengths to correct (D-02); the unanchored stand tips at 16.5 m/s, so anchoring is required (D-09)	High	Plate tongue bends need care; the zigzag edges are hand-cut	SSK-BLD-001; `cad/fidelity.json`
Power and controls	None: passive, gravity feed from a container at most 1.5 m high	High	Feed must be refilled daily; a dry day drives stagnation	PR-005; SSK-CAL-001 section 5
Supply of parts	Common building, metal and plastics supplies; PPS strip and a certified coater are the two specialist items	Medium	PPS priced from a distributor list price, not a quote; coater availability by region unknown	BOM lines 5 and 6; SSK-DEC-001 to confirm items 5, 7, 8
Cost	USD 460.70 per 1 m2 panel, USD 105.70 over the value-engineering target	Low	Coated plates and PPS rails dominate; savings listed in SSK-DEC-001	SSK-CAL-001 section 10